System Realization and Filter Design By Poles and Zeros Positioning

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Outline

- Laplace Transform
- Properties of Laplace Transform
- Solution of Differential Equations
- Analysis of Electrical Networks
- Frequency Response of an LTIC System
- Block Diagrams and System Realization
- Filter Design by Placement of Poles and Zeros of $H(s)$

Transfer Function

Time Delay

$x(t)$ → **Time Delay τ** → $y(t) = x(t-\tau)$

$X(s)$ → **Time Delay τ** → $Y(s) = X(s)e^{-\tau s}$

Shifting Property

$$\boldsymbol{H(s) = e^{-\tau s}}$$

Differentiator

$x(t)$ → **Differentiator** → $y(t) = \dfrac{dx(t)}{dt}$

$X(s)$ → **Differentiator** → $Y(s) = s\,X(s)$

Differentiator Property

$$\boldsymbol{H(s) = s}$$

Integrator

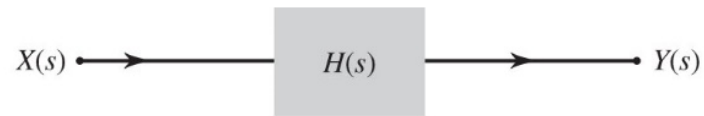
$x(t)$ → **Integrator** → $y(t) = \int x(t)dt$

$X(s)$ → **Integrator** → $Y(s) = \dfrac{1}{s}X(s)$

Integrator Property

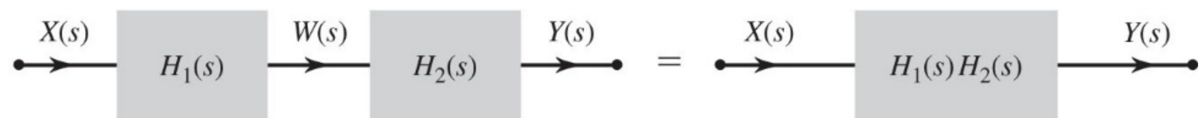
$$\boldsymbol{H(s) = \frac{1}{s}}$$

Block Diagrams



(a)

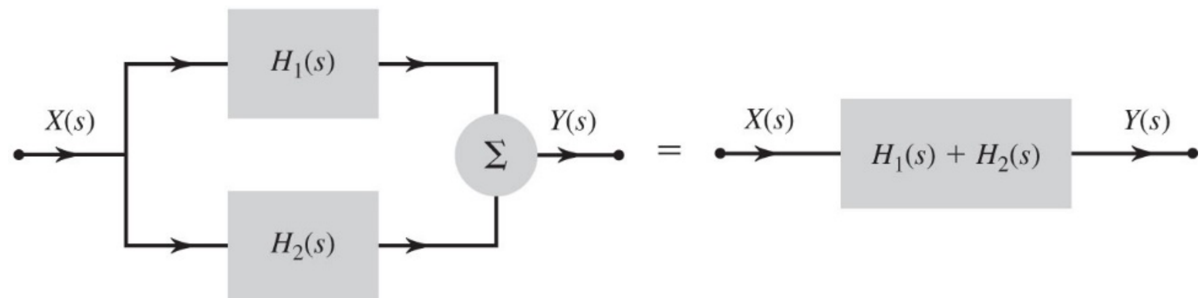
Cascade Connection



(b)

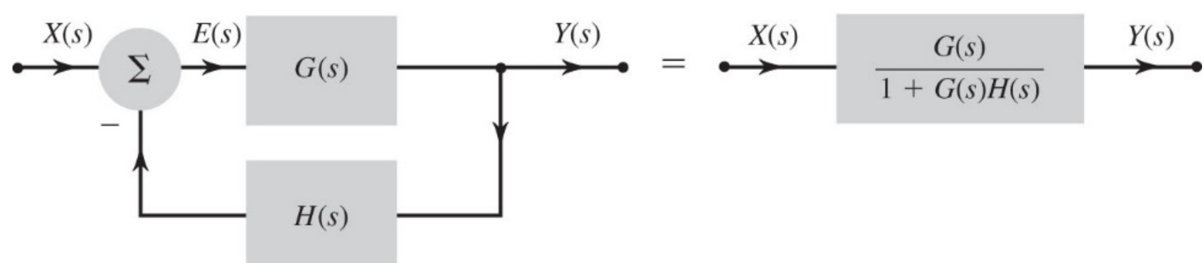
$$\frac{Y(s)}{X(s)} = \frac{W(s)}{X(s)}\frac{Y(s)}{W(s)} = H_1(s)H_2(s)$$

Parallel Connection



(c)

Feedback Interconnections



(d)

System Realization

$$H(s) = \frac{b_0 s^M + b_1 s^{M-1} + \ldots + b_M}{s^N + a_1 s^{N-1} + \ldots + a_N}$$

- Realization is a synthesis problem, so there is no unique way of realizing a system.
- A common realization of $H(s)$ is using
  - Integrator
  - Scalar multiplier
  - Adders

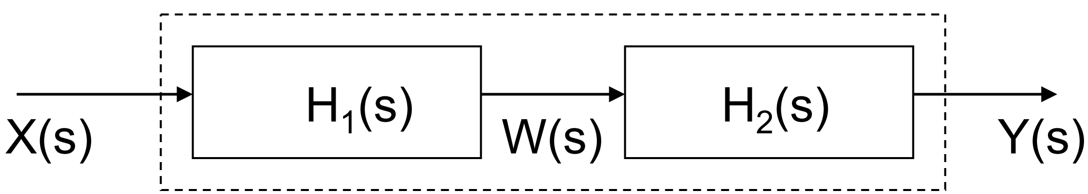
Direct Form I Realization

$$H(s) = \frac{b_0 s^3 + b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3}$$

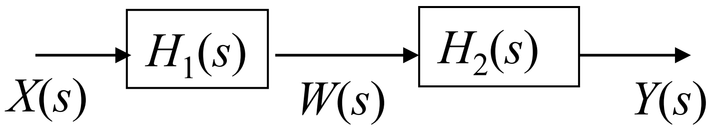
Divide every term by s with the highest order in the denominator, s^3.

$$H(s) = \frac{b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3}}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}}$$

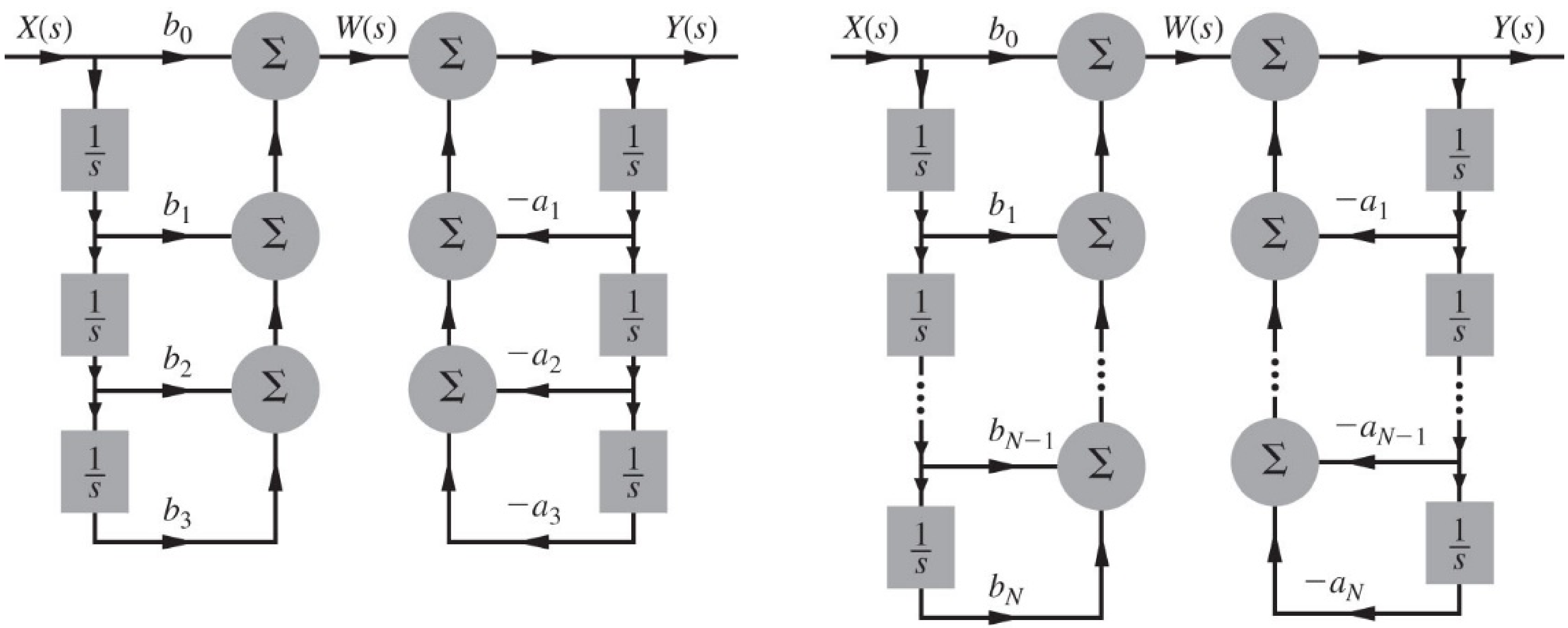
$$H(s) = \overbrace{\left(b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3}\right)}^{H_1(s)} \overbrace{\left(\frac{1}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}}\right)}^{H_2(s)}$$



Direct Form I Realization

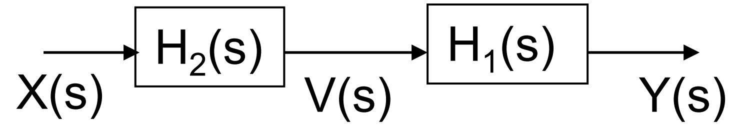


$$H(s) = \left(b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3}\right)\left(\frac{1}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}}\right)$$

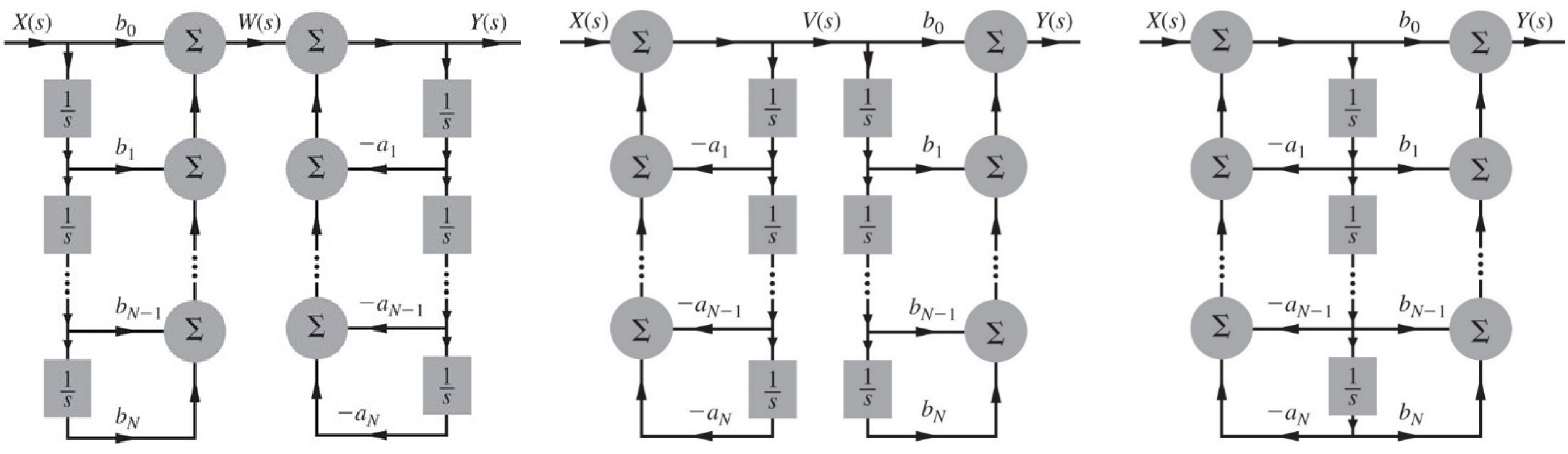


(a) (b)

Direct Form II Realization



$$H(s)=\left(\frac{1}{1+\frac{a_1}{s}+\frac{a_2}{s^2}+\frac{a_3}{s^3}}\right)\left(b_0+\frac{b_1}{s}+\frac{b_2}{s^2}+\frac{b_3}{s^3}\right)$$



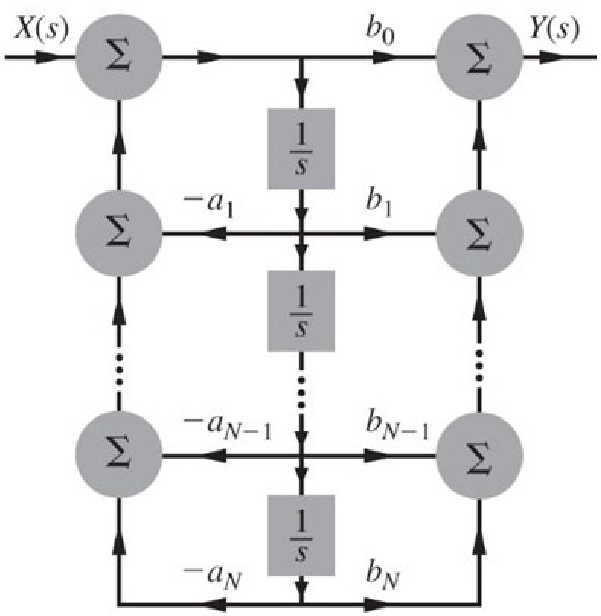
(a)

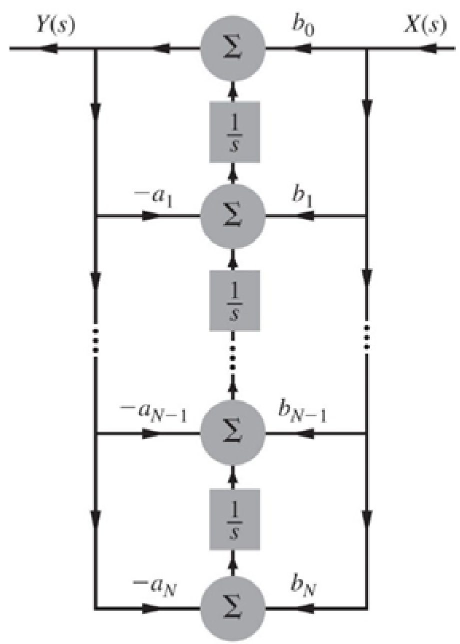
(b)

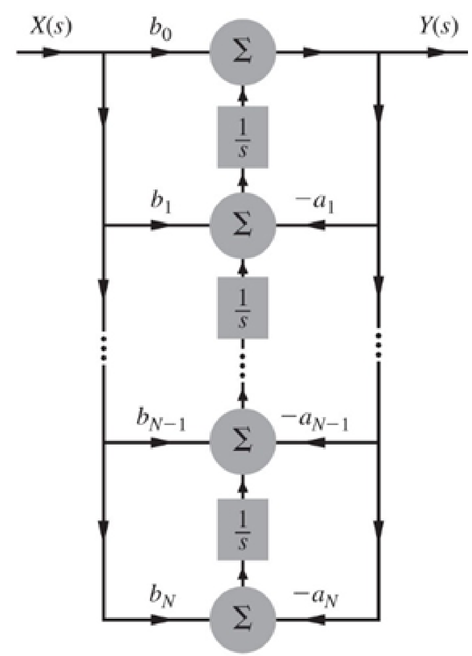
Transposed Realization

- Reverse all the arrow directions without changing the scalar multiplier values.
- Replace branch nodes by adders and vice versa.
- Replace the input X(s) with the output Y(s) and vice versa.
- To make the follow of the input from left to right rotate the diagram 180 degree.

$$H(s) = \left(b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3}\right)\left(\frac{1}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}}\right)$$







Example

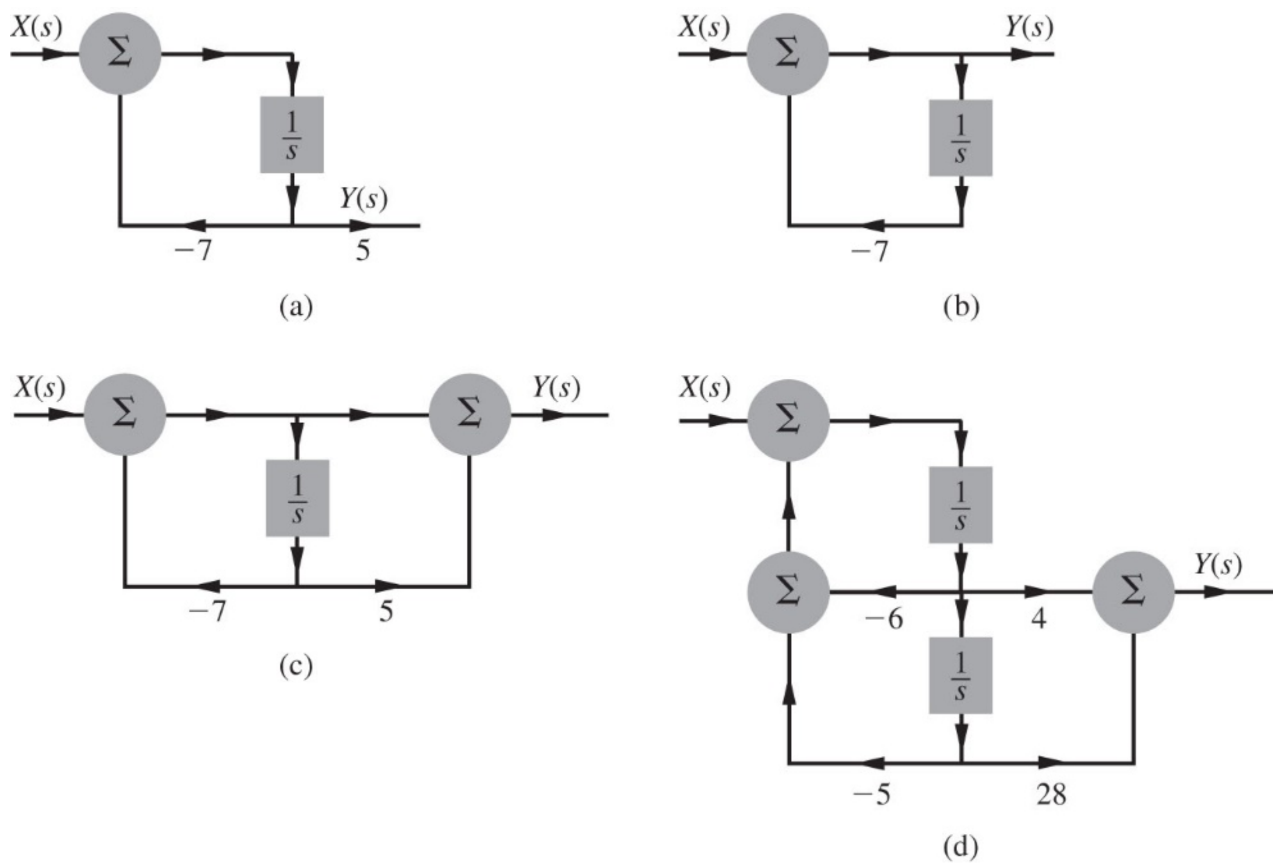
Find the canonic direct form realization of the following transfer functions:

a) $\dfrac{5}{s+7}$

b) $\dfrac{s}{s+7}$

c) $\dfrac{s+5}{s+7}$

d) $\dfrac{4s+28}{s^2+6s+5}$



Cascade and Parallel Realizations

$$H(s) = \frac{4s+28}{s^2+6s+5}$$

Cascade Realization

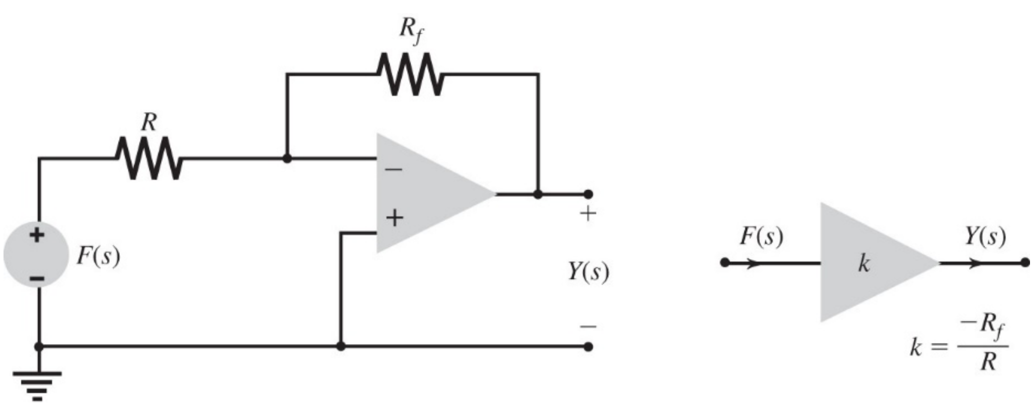
$$H(s) = \frac{4s+28}{(s+1)(s+5)} = \left(\frac{4s+28}{s+1}\right)\left(\frac{1}{s+5}\right)$$

Parallel Realization

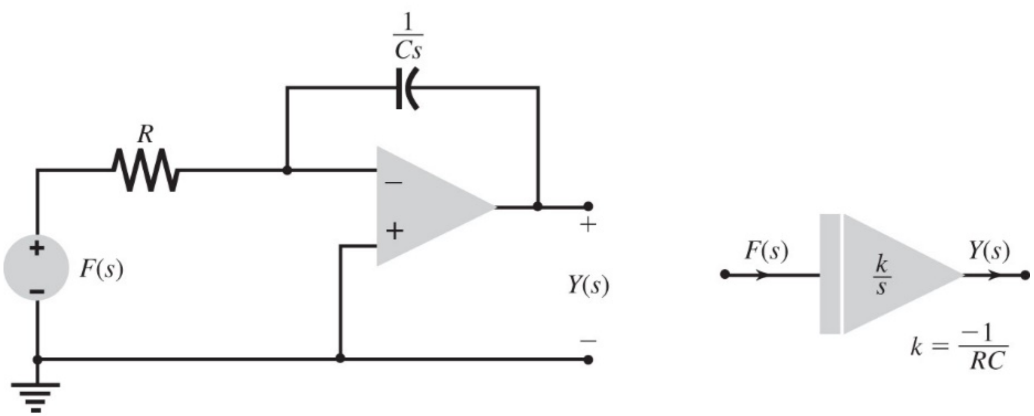
$$H(s) = \frac{4s+28}{(s+1)(s+5)} = \frac{6}{s+1} - \frac{2}{s+5}$$

The complex poles in $H(s)$ should be realized as a second-order system.

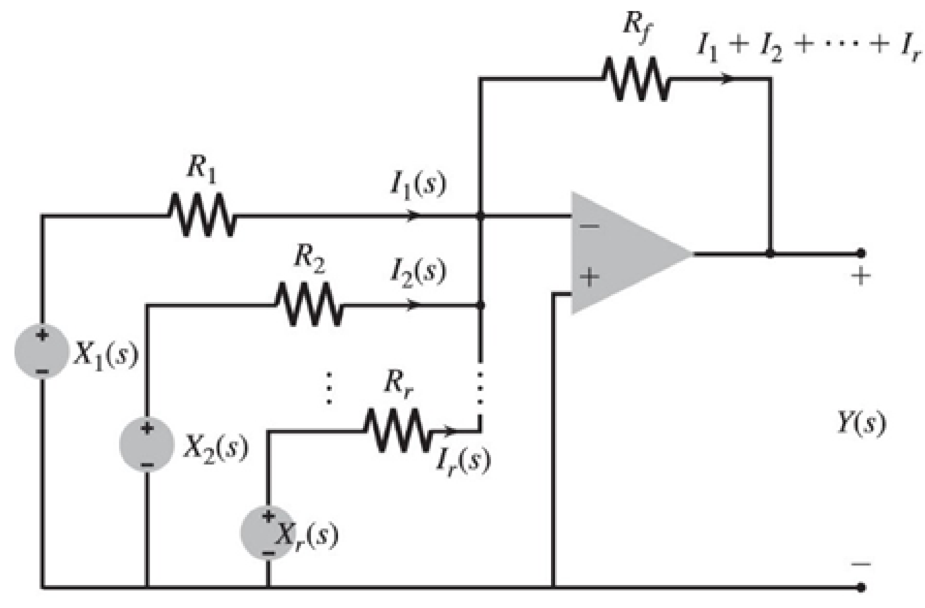
Using Operational Amplifier for System Realization



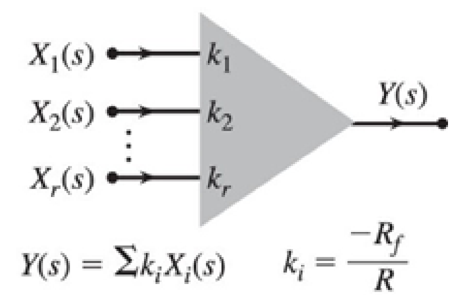
(a)



(b)



(a)

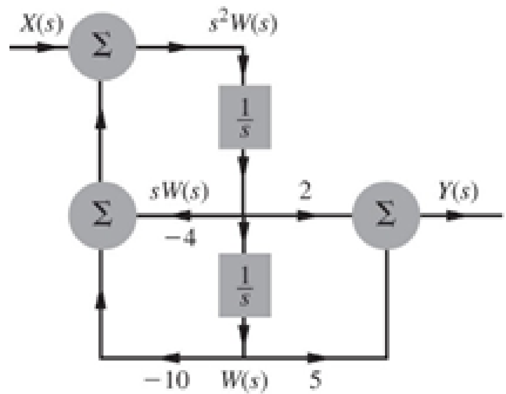


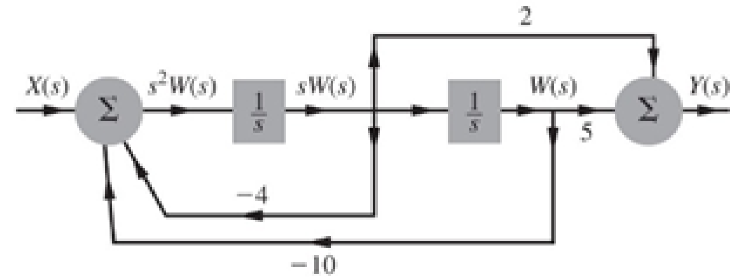
(b)

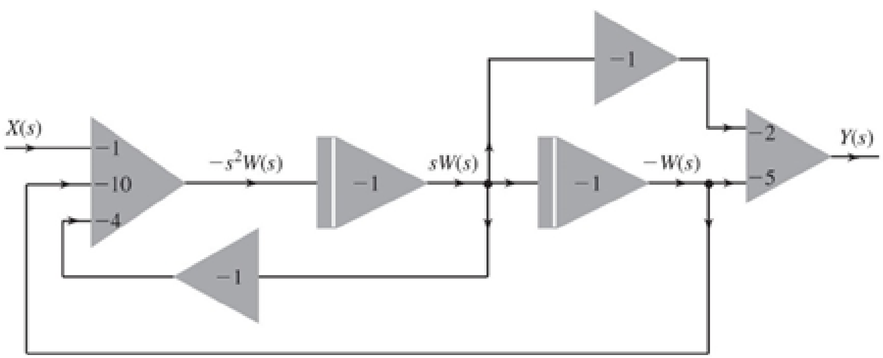
Example

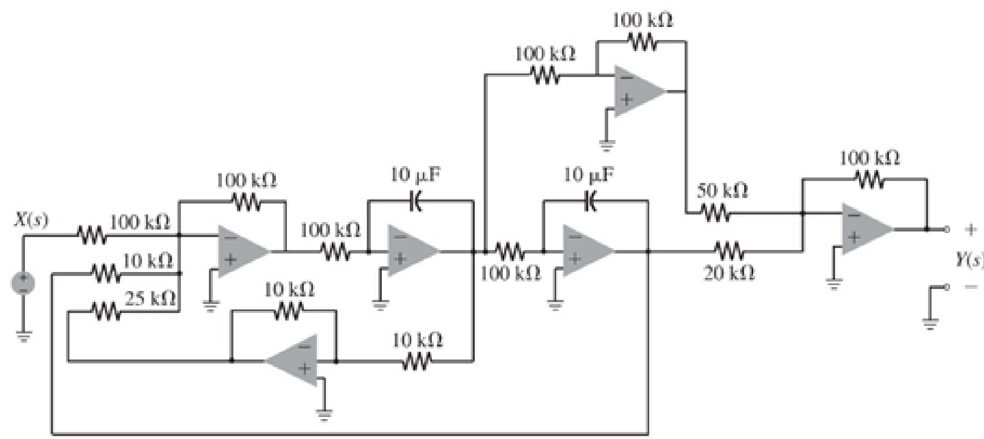
Use Op-Amp circuits to realize the canonic direct form of the transfer function

$$H(s)=\frac{2s+5}{s^2+4s+10}$$







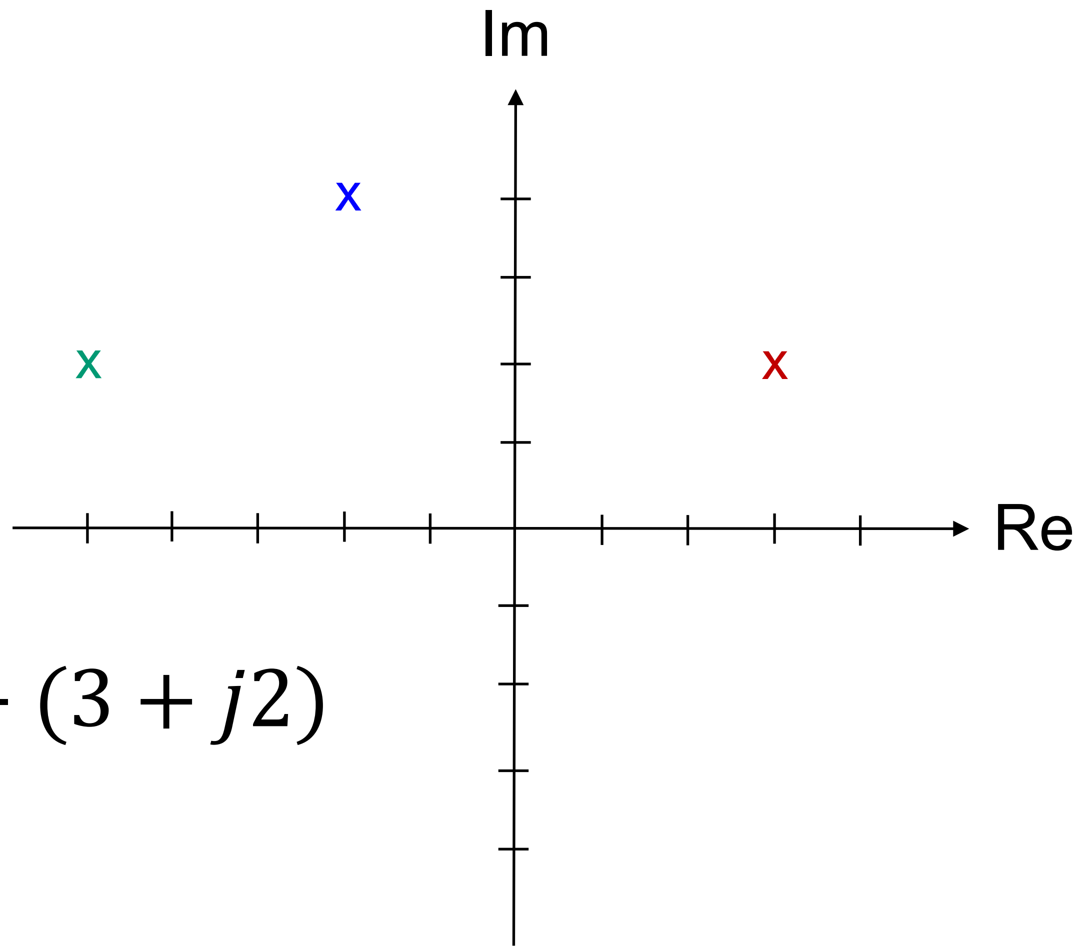


Filter Design By Poles and Zeros Positioning

Review of Vectors Operations

$\boldsymbol{A} = -2 + j4$

$\boldsymbol{B} = 3 + j2$



C = A - $\boldsymbol{B}$ $= (-2 + j4) - (3 + j2)$

C = A - $\boldsymbol{B}$ $= -5 + j2$

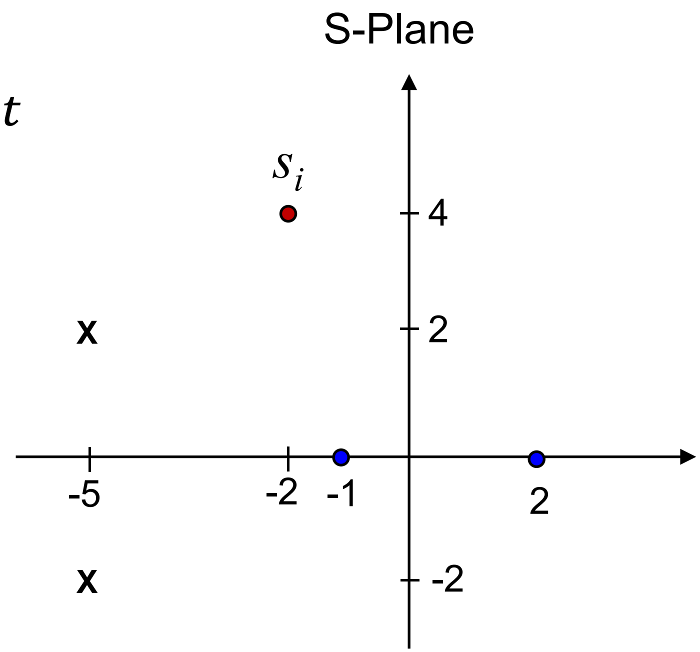
D = A$\boldsymbol{B}$

D = A/$\boldsymbol{B}$

Effects of Poles & Zeros on Frequency Response

$$s_i = -2+j4$$

$$x(t) = e^{s_i t} \longrightarrow \boxed{H(s)} \longrightarrow y(t) = H(s_i)e^{s_i t}$$



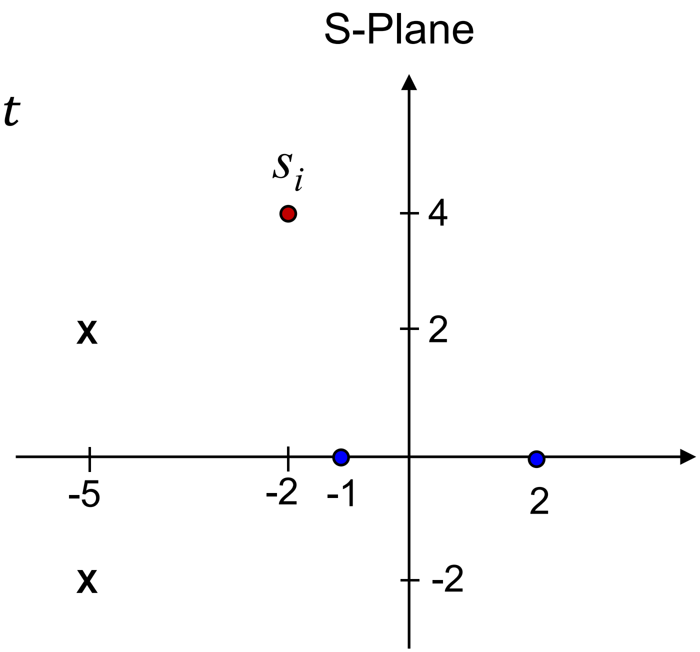
$$H(s) = \frac{s^2 - s - 2}{s^2 + 10s + 29}$$

$$H(s) = \frac{(s-2)(s+1)}{(s+5-j2)(s+5+j2)}$$

Effects of Poles & Zeros on Frequency Response

s_i = -2+j4

$$x(t) = e^{s_i t} \longrightarrow \boxed{H(s)} \longrightarrow y(t) = H(s_i)e^{s_i t}$$



$$H(s) = \frac{s^2 - s - 2}{s^2 + 10s + 29}$$

$$H(s) = \frac{(s-2)(s+1)}{(s+5-j2)(s+5+j2)}$$

$$H(s_i) = \frac{(-2+j4-2)(-2+j4+1)}{(-2+j4+5-j2)(-2+j4+5+j2)} = \frac{(-4+j4)(-1+j4)}{(3+j2)(3+j6)}$$

$$H(s_i) = \frac{(\sqrt{32}e^{j135})(\sqrt{17}e^{j104})}{(\sqrt{13}e^{j33.7})(\sqrt{45}e^{j63.4})} = 0.96e^{j141.9}$$

Effects of Poles & Zeros on Frequency Response

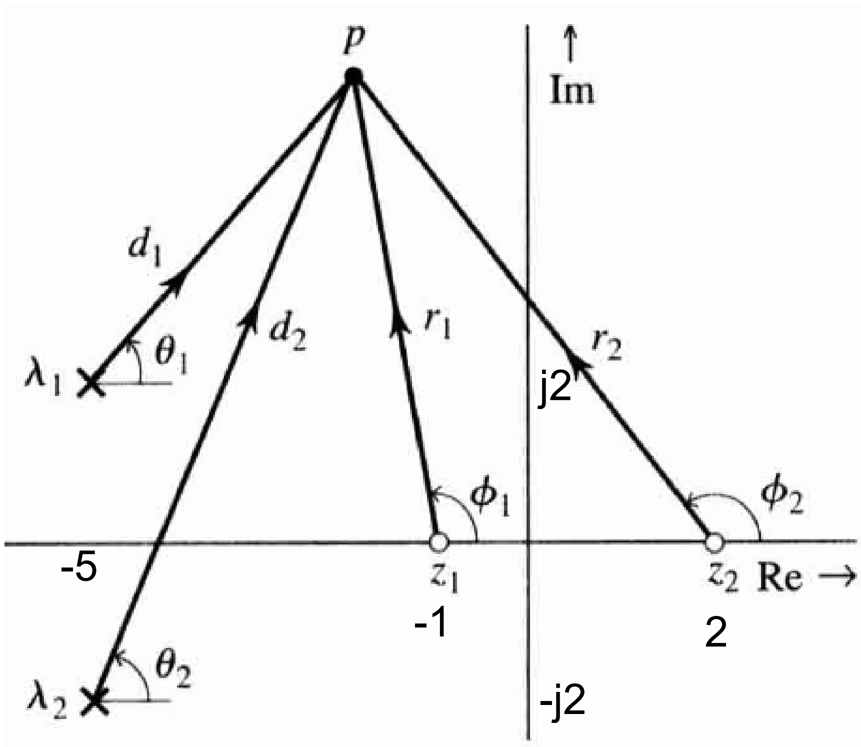
Therefore the magnitude and phase at $s = p$ are given by:

$$H(s) = \frac{s^2 - s - 2}{s^2 + 10s + 29}$$

$$H(s) = \frac{(s-2)(s+1)}{(s+5-j2)(s+5+j2)}$$

$$|H(s)|_{s=p} = b_0 \frac{r_1 r_2 \cdots r_N}{d_1 d_2 \cdots d_N}$$

$$= b_0 \frac{\text{product of the distances of zeros to } p}{\text{product of the distances of poles to } p}$$



$$\angle H(s)|_{s=p} = (\phi_1 + \phi_2 + \cdots + \phi_N) - (\theta_1 + \theta_2 + \cdots + \theta_N)$$

$$= \text{sum of zero angles to } p - \text{sum of pole angles to } p$$

Effects of Poles & Zeros on Frequency Response

Consider a general system transfer function:

$$H(s) = \frac{P(s)}{Q(s)} = b_0 \frac{(s - z_1)(s - z_2)\cdots(s - z_N)}{(s - \lambda_1)(s - \lambda_2)\cdots(s - \lambda_N)}$$

zeros at $z_1, z_2, \ldots, z_N$

poles at $\lambda_1, \lambda_2, \ldots, \lambda_N$

The value of the transfer function at some complex frequency $s = p$ is:

$$H(s)|_{s=p} = b_0 \frac{(p - z_1)(p - z_2)\cdots(p - z_N)}{(p - \lambda_1)(p - \lambda_2)\cdots(p - \lambda_N)} = b_0 \frac{(r_1 e^{j\phi_1})(r_2 e^{j\phi_2})\cdots(r_N e^{j\phi_N})}{(d_1 e^{j\theta_1})(d_2 e^{j\theta_2})\cdots(d_N e^{j\theta_N})}$$

$$|H(s)|_{s=p} = b_0 \frac{r_1 r_2 \cdots r_N}{d_1 d_2 \cdots d_N} = b_0 \frac{\text{product of the distances of zeros to } p}{\text{product of the distances of poles to } p}$$

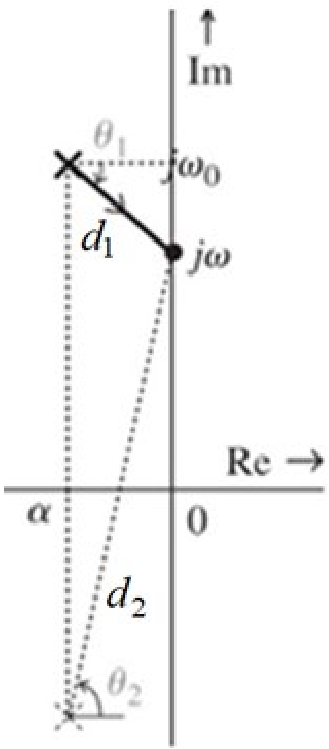
$$\begin{aligned} \angle H(s)|_{s=p} &= (\phi_1 + \phi_2 + \cdots + \phi_N) - (\theta_1 + \theta_2 + \cdots + \theta_N) \\ &= \text{sum of zero angles to } p - \text{sum of pole angles to } p \end{aligned}$$

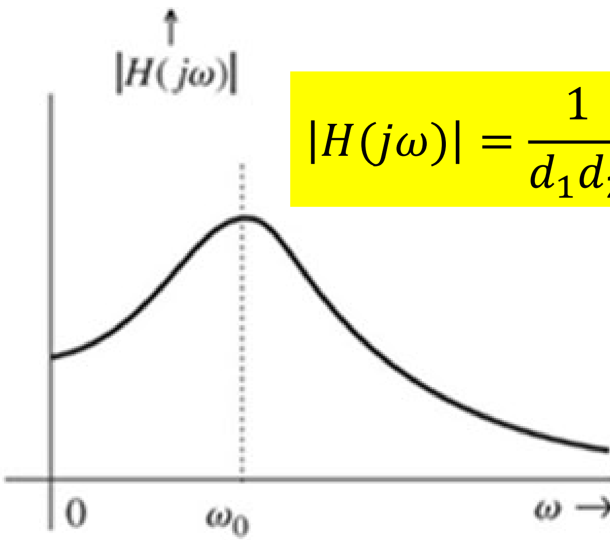
Poles Enhance Amplitude Frequency Response

- Frequency Response of a system is obtained by evaluating $H(s)$ along the y-axis (i.e. taking all value of $s = j\omega$).

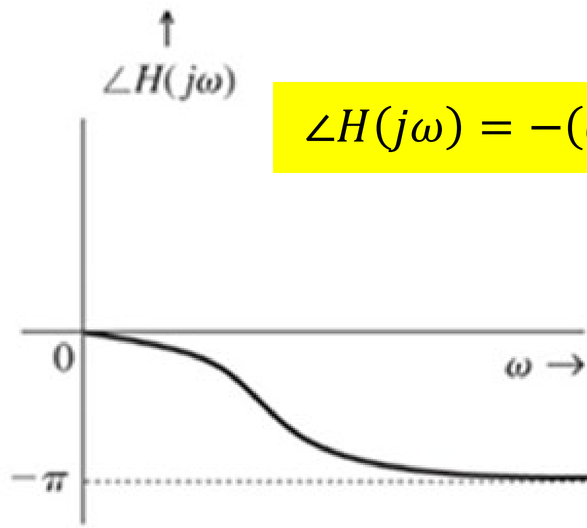
- Consider the effect of two complex system poles on the frequency response.

$$H(s) = \frac{1}{(s-(\alpha + j\omega))(s-(\alpha - j\omega))}$$





$$|H(j\omega)| = \frac{1}{d_1 d_2}$$

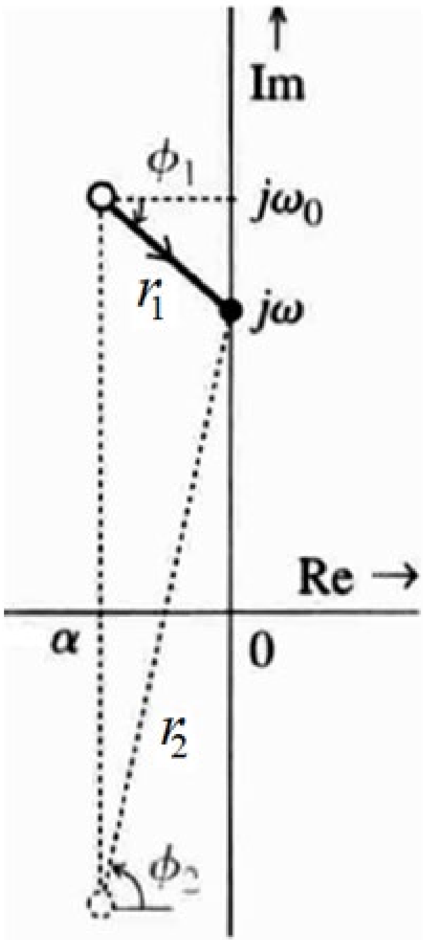


$$\angle H(j\omega) = -(\theta_1 + \theta_2)$$

Systems amplify frequencies near a pole

Zeros Suppresses Amplitude Frequency Response

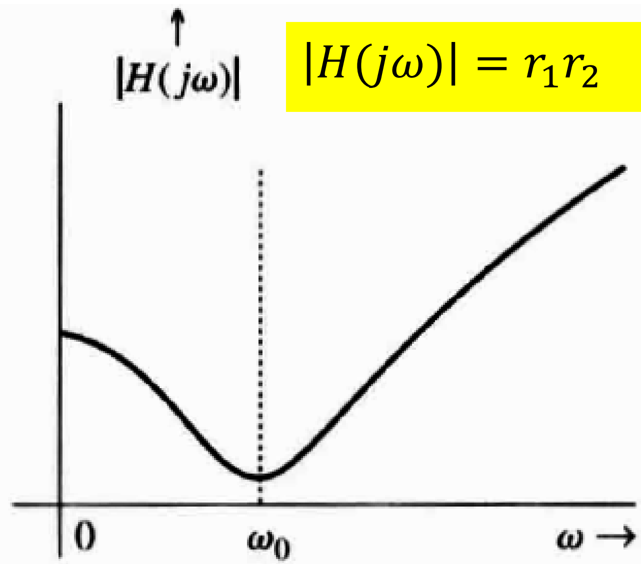
Consider the effect of two complex system zeros on the frequency response.

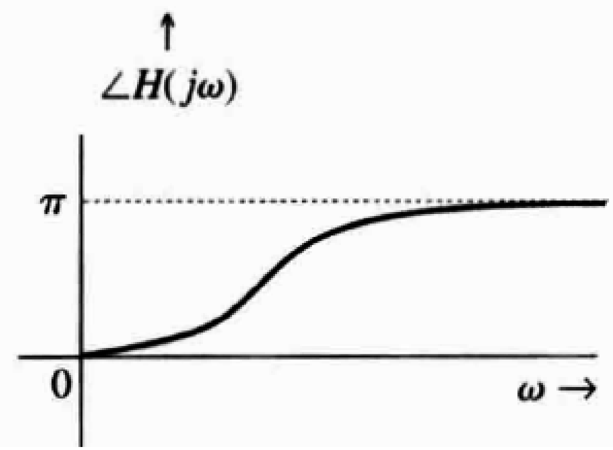


$$H(s) = (s - (\alpha + j\omega))(s - (\alpha - j\omega))$$

$$|H(j\omega)| = r_1 r_2$$

$$\angle H(j\omega) = -(\emptyset_1 + \emptyset_2)$$





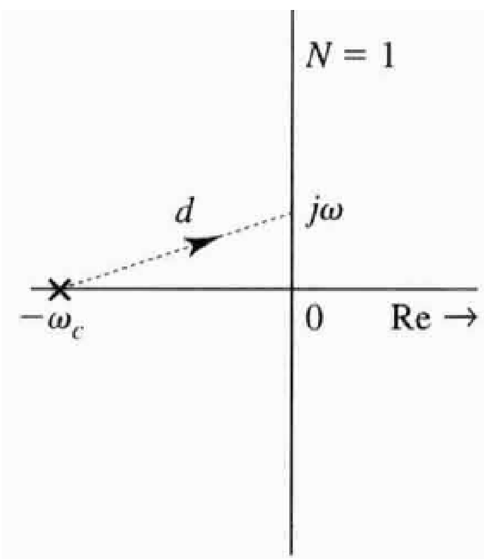
Systems suppress frequencies near a zero

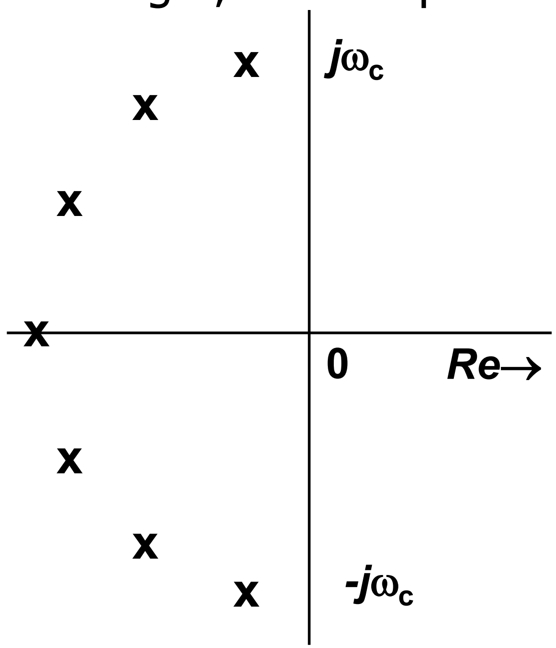
Low-pass Filters

- Use the enhancement and suppression properties of poles & zeros to design filters.
- Low-pass filter (LPF) has maximum gain at $\omega=0$, and the gain decreases as ω increase.
- Simplest LPF has a single pole on real axis, say at ($s=-\omega_c$), then

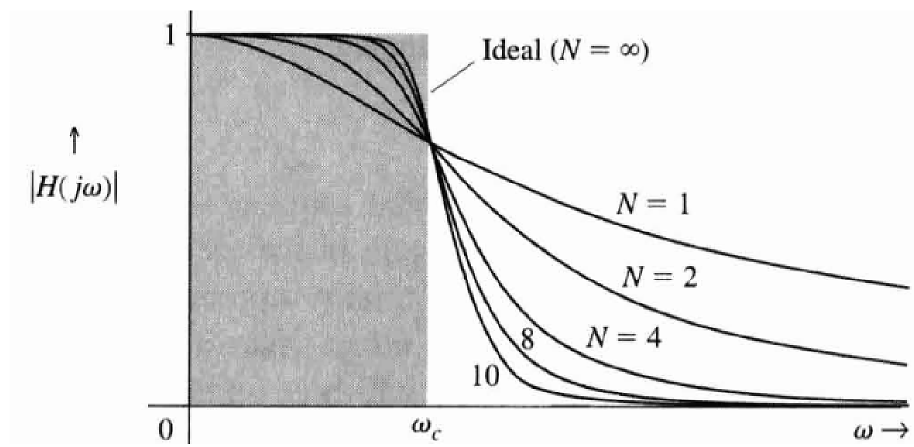
$$H(s) = \frac{\omega_c}{s+\omega_c} \quad \text{and} \quad |H(j\omega)| = \frac{\omega_c}{d}$$

- To have an ideal LPF (i.e. very sharp cut-off at ω_c), we need a WALL OF POLE as shown, the more poles we get, the sharper the cut-off.



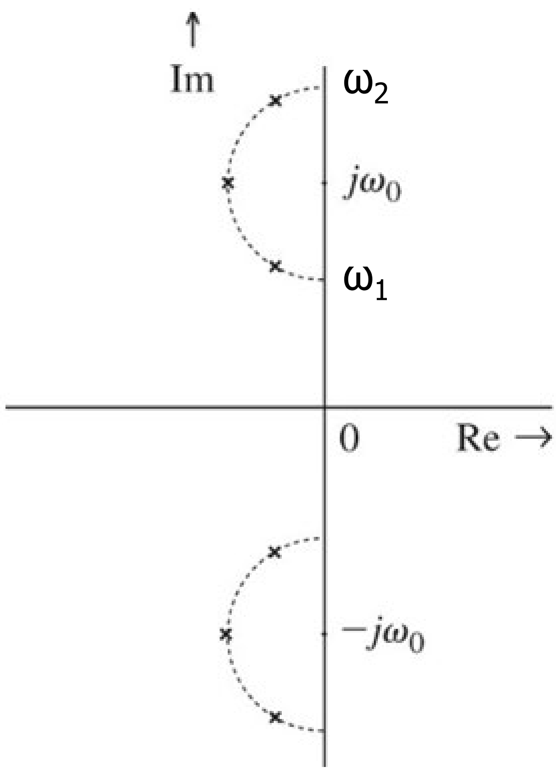


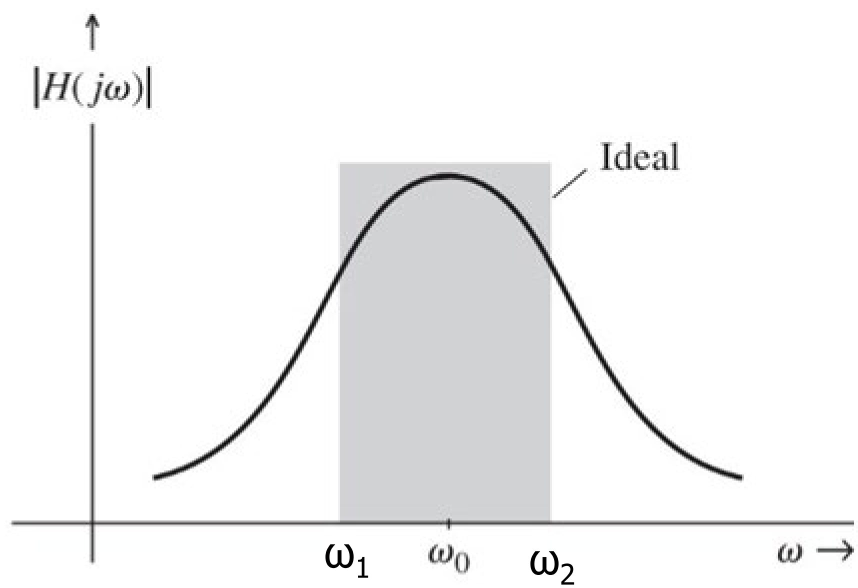
Butterworth Filter



Band-pass Filter

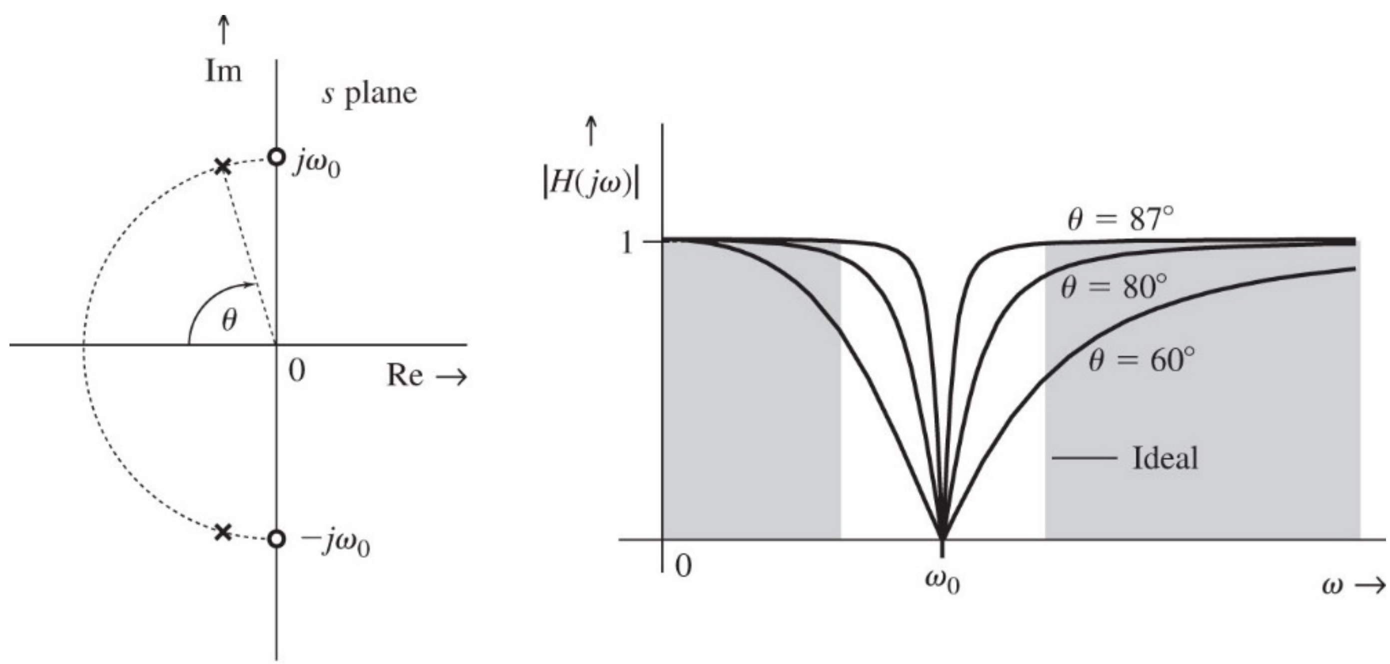
- Band-pass filter has gain enhanced over the entire passband, but suppressed elsewhere.
- For a passband centred around ω_0, we place many poles in a semicircle where its center is ω_0 and diameter equals the difference between the upper ω_2 and lower cutoff frequencies ω_1.





Notch Filter

- A notch filter suppresses specific frequency component such as ω_0 in a signal and allows other frequencies to pass without attenuation. To suppress ω_0, two zeros are placed in $\pm j\omega_0$.
- To allow other frequencies to pass without attenuation two poles should be placed near the two zeros in the semicircle in left hand side plane.



(a) (b)

Notch Filter Example

Design a second-order notch filter to suppress 60 Hz hum in a radio receiver.

$$H(s) = \frac{(s - j\omega_o)(s + j\omega_o)}{(s + \omega_o cos\theta + j\omega_o sin\theta)(s + \omega_o cos\theta - j\omega_o sin\theta)}$$

$$= \frac{s^2 + \omega_o^2}{s^2 + (2\omega_o cos\theta)s + \omega_o^2}$$

$$= \frac{s^2 + 142122.3}{s^2 + (754 cos\theta)s + 142122.3}$$